

SENIOR DESIGN MINI PROJECT 3

CHE 396

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Contents

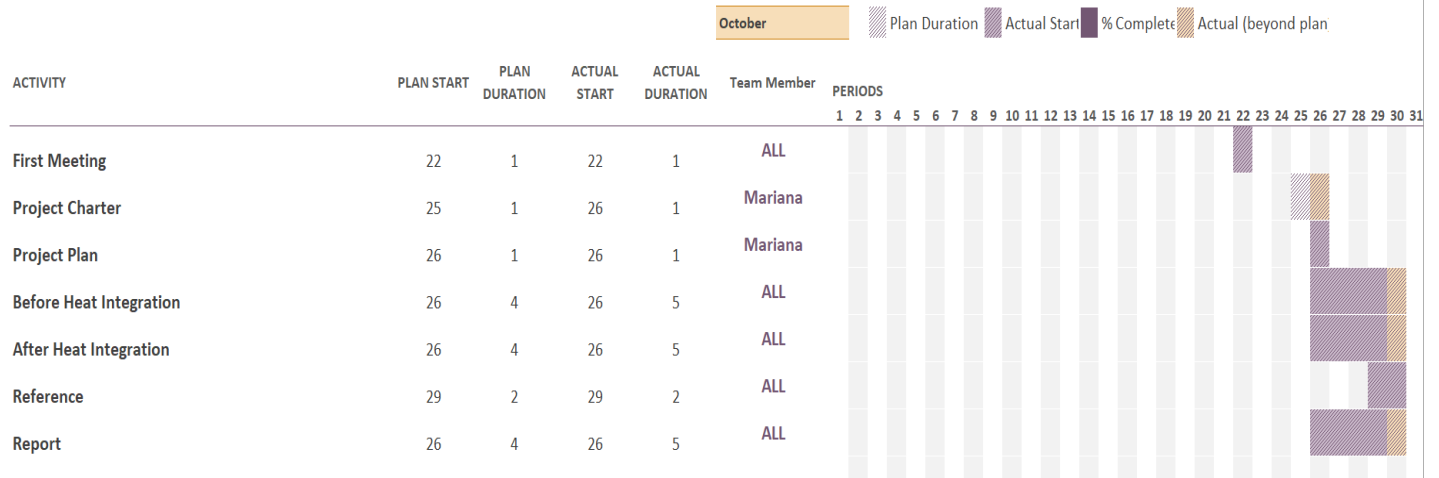
1.0 Project Charter	3
2.0 Project Plan	4
3.0 Before Heat Integration	5
3.1 Left-Hand Side Before heat Integration	5
3.2 Right-Hand Side Before Heat Integration	10
4.0 After Heat Integration.....	14
4.1 Left-Hand Side Before heat Integration	15
4.2 Right-Hand Side Before Heat Integration.....	19
5.0 Conclusion.....	24
6.0 References	25

1.0 Project Charter

Title	Goal	In Scope	Out of Scope
Mini Project 3: Production of vinyl chloride in an efficient and safely manner.	Th goal of this project is to produce vinyl chloride in an efficient and safely manner, by choosing the right construction material for the equipment and pipes, and by designing a control system.	Design of control system Manipulated variables Choose the equipment's material Create a safe and efficient process to produce vinyl chloride	Potential Disturbances Identify any Constrains Discuss the Process
Deliverable		Team Members	
- Material of Construction - Piping and Instrumentation Diagram - Conceptual Control System		<i>Erica Caruso</i> <i>Yaser Khalid</i> <i>Mariana St.Peter</i>	

2.0 Project Plan

Project Planner



3.0 Before Heat Integration

In this section it will be presented the production of vinyl chloride before the heat integration takes place. The analysis will be divided between the left and right side of the process.

3.1 Left Side of Process

Figure 1 below shows the left side of the Vinyl Chloride production before heat integration. This portion of the process contains a direct chlorination reactor, a pyrolysis furnace and a quench tank. It also contains streams 1 to 8, which will be further analyzed in order to determine the construction material for the system.

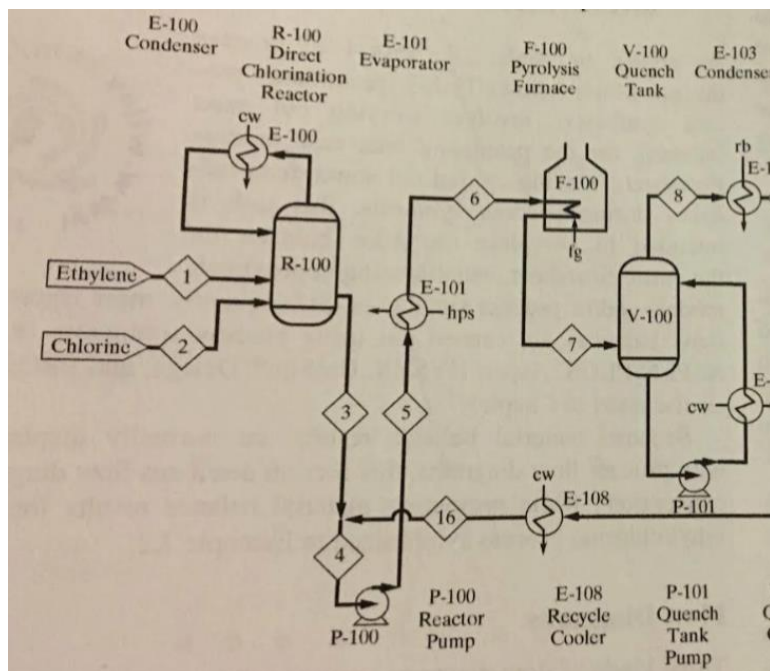


Figure 1: Left side of vinyl chloride production before heat integration

3.1.1 Materials of Construction

Table 1: Properties of streams 1 to 8.

	Stream 1	Stream 2	Stream 3	Stream 4	Stream 5	Stream 6	Stream 7	Stream 8
Temp (°C)	25	25	90	90	91.3	242	500	170
Pressure (atm)	1.5	1.5	1.5	1.5	26	26	26	26
Molar flow (lbmol/hr)								
Ethylene	1600	0	0	0	0	0	0	0
Chlorine	0	1600	0	0	0	0	0	0
1,2 - dichloroethane	0	0	1600	2667	2667	2667	1067	1067
Vinyl chloride	0	0	0	0	0	0	1600	1600
Hydrogen chloride	0	0	0	0	0	0	1600	1600

In order to choose which material to use for the left side of the process, each stream must be analyzed based on its compositions and physical properties, such as temperature and pressure.

Stream 1 is the ethylene fresh feed stream; its composition is made solely by ethylene at 1.5 atm and 25°C. Stream 2 is the chloride fresh feed; its composition is made solely by chlorine at 1.5 atm and 25°C.

Stream 3 is the resultant stream after the chemical reaction takes place. Its composition is made by 100% 1,2-dichloroethane at 90°C and 1.5 atm. Stream 4 has the same composition as stream 3 at the same temperature and pressure, however, with a higher flow rate since it is a combination of stream 3 and stream 16. Stream 5 has the same composition as stream 3 and 4, however, since this stream is an outlet of a pump, its pressure is 26 atm at 91.3°C. Stream 6 is the resultant stream of a heating process of stream 5, which means its pressure is maintained at 26 atm, but its temperature is brought up to 242°C. Since no chemical reactions occurred, stream 6 has the same composition and flow rate as stream 5.

Finally, stream 6 is fed to the pyrolysis furnace where a thermal cracking takes place and a chemical reaction occurs. Stream 7 is the result of the pyrolysis with a composition of 1,2-dichloroethane, vinyl chloride and hydrogen chloride at 500°C and 26 atm, which is then fed to the quench tank, lowering its temperature to 170°C and retaining the same composition and pressure as stream 7.

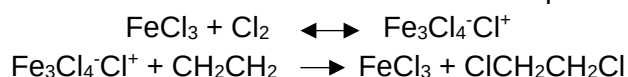
The focus of the left side of the process is the three major equipment presented in this part: two reactors and the furnace. Based on research, carbon steel is used to fabricate the shell of the first reactor [1]. Dry chlorine at room temperature doesn't present high risk of corrosion and can be transported and handled in carbon steel containers or pipes, while "wet chlorine is extremely corrosive to steel and many nickel alloys and requires Allow C-276 or titanium" [2]. Vessels made out of carbon steel are usually limited to temperatures between 120°C – 150°C.

Even though stainless steel is more resistant than carbon steel, SS is not used due to the possibility of moisture during possible shutdown, which would consequence in stress corrosion cracking [2]. As a result, carbon steel is the choice of material for the reactor and auxiliary equipment, as long as the feedstock is dried, and the temperature is controlled in a proper manner.

For the pyrolysis furnace, carbon brick is the choice of material, due to its high temperature properties and to the best acid resistance properties when compared with other types of ceramic using hydrochloric acid [3]. Tubing must be in stainless steel due to the production of hydrochloric acid. Stainless steel material can handle HCl as long as the acid is totally dried, which explains the high temperatures of the streams carrying the component after the pyrolysis furnace. Carbon steel would have almost the same corrosion characteristics toward the hydrochloric dried acid, however, carbon steel is not designed to as a high temperature as of the streams. This also results in stainless steel for the second reactor for the same reasons discussed.

3.1.2 Constrains

Chemical reaction inside the reactor in the presence of ferric chloride catalyst forming 1,2 – dichloroethane by reacting ethylene and chlorine. Reactor temperature shall not be higher than 100°C. Since this process is exothermic, $\Delta H = -180$ kJ/mol, cooling water must be used to remove the heat of reaction [1]. Chemical reaction follows the balanced equation:



In the pyrolysis furnace occurs the production of vinyl chloride, which is an endothermic reaction $\Delta H = 71$ kJ/mol [1]. Furnace is composed of four sections, the radiation section, where tubes, burners, tube sheets can be found at 500°C. The convection section, where the feed is preheated. The shock and stack, where gases from the combustion are treated and released. Furnace temperature shall be controlled by the fuel gas, which is a ratio control of effluent flowrate and temperature dependent [1].

3.1.3 Control Objectives

Table 2: Control Variables for left side of the process

Equipment	Controlled Variable	Manipulated Variable
Feed flowrate	Chlorine flowrate	Ethylene Flowrate
1 st Reactor	Temperature of reactor	Colling water flowrate
1 st Reactor	Pressure of Reactor	Safety Pressure release Valve
Pyrolysis Furnace	Inside temperature of the furnace	Fuel gas flowrate
2 nd Reactor	Temperature of the Reactor	Cooling water flowrate

As Table 2 indicates, there are a couple-controlled variable in the left side of the process. In addition to those variables listed above, in the process was added two extra controls for the feed, which will control the ratio going into the reactor. In case of an increase of ethylene fresh feed, the flow transmitter would send a signal to the flow indicator control, which would control the flow valve to adjust the flowrate. The system then would respond by increasing the flowrate of chlorine. The same would happen for decrease of fresh feed.

For the first reactor there are two controlled variables available. Since it is extremally important to keep the temperature under control inside the reactor, the vessel has a cooling jacket around it. The fluid inside the jacket is then cooled down by a cooling water stream. A temperature element connected to the reactor will transmit the signal to a temperature transmitter, which will inform the temperature control to change the flow of the cooling water by rotating the temperature valve, which is responsible for the cooling water flowrate. In case the system needs to cool down further, the water flowrate would increase, cooling the jacket fluid to the desired temperature. In case there is no need for such a low temperature, the system would rotate the valve limiting the flowrate of the cooling water, which would result in a higher temperature of the jacket's fluid. The second controlled variable for the reactor is a safety pressure released valve. It is known from thermodynamics that the pressure of a vessel can be controlled by controlling its temperature. For a moment it seems unnecessary the addition of an extra control, however, as the same indicates it is a safety pressure release valve. This valve will only be activated in case of something went wrong with the temperature control, as a last resource to avoid a catastrophic explosion from pressure build up inside the vessel. In the case of the temperature control fails, the pressure transmitter sends a signal to the pressure indicator control, which will operate the pressure valve to release the extra pressure.

Pyrolysis furnace only has one controlled variable, temperature. To reach the desired temperature inside the furnace, the temperature element will send measurements to the temperature transmitter, which is connected with the temperature control, by sending a signal to the temperature valve, the flowrate of fuel gas will be adjusted accordantly with the needs of the furnace. By increasing the flowrate of fuel gas, the temperature will increase, and the same is true for decreasing the fuel gas flowrate will result in a decrease of temperature inside the furnace.

The second reactor, which works as a quench tank also only has one manipulated variable, its temperature. To obtain the outlet stream at the desired temperature, a cooling water stream is used for the recycling material. A temperature element is connected to the stream coming out of the top of the reactor. The temperature element then sends a signal to the temperature transmitter, which communicates to the temperature control on how to operate the temperature valve responsible for the cooling water flow. By increasing the flow of the cooling water, the vessel internal temperature will decrease, which will result in a colder outlet stream. On the other hand, if the cooling water flowrate decreases, the internal temperature of the tank will increase, which will result in a higher temperature for the outlet stream. A visual summary of this discussion can be seen at Figure 2.

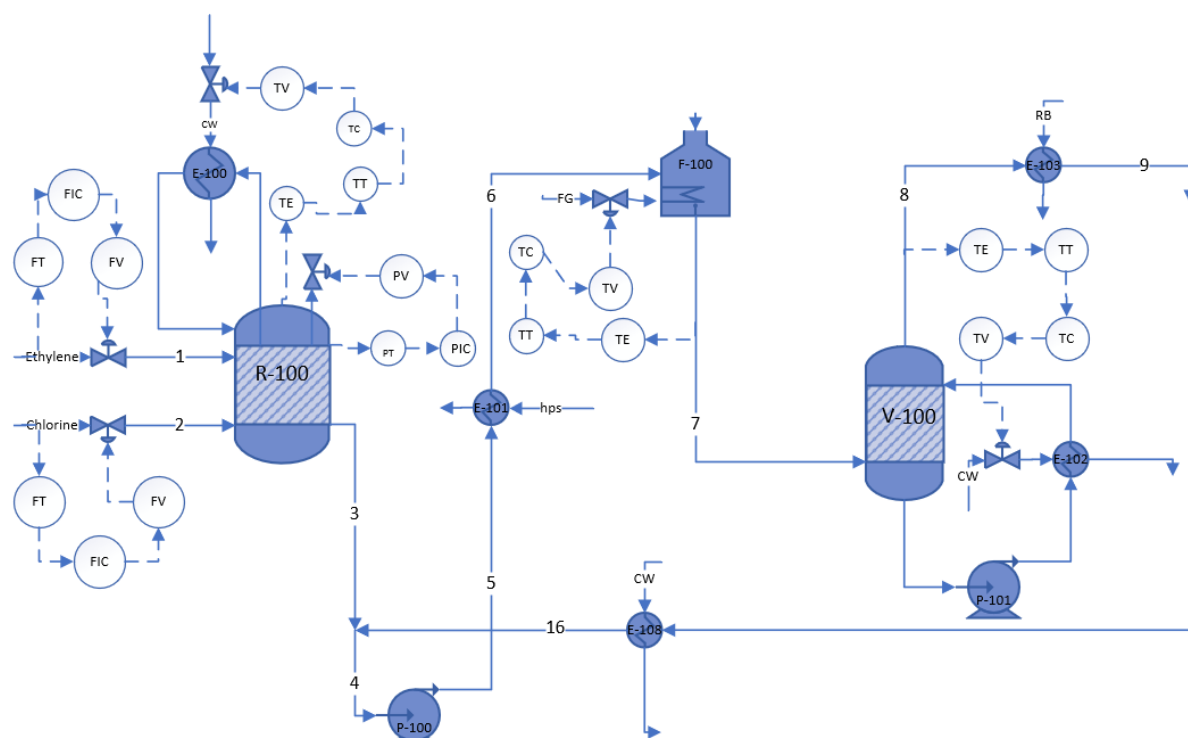


Figure 2: Process Control of left side of the Process

3.1.4 Safe Limits for Process Variables

Safe limits of operation would be the first reactor to not exceed 120°C and 15 psig [1]. Carbon steel is not designed to undergo high temperatures. After hydrochloric acid is formed, system must dry out the acid to avoid any corrosion in pipes and equipment.

Feed flowrate must be compatible with reactor size to not overflow the reactor and waste fresh material.

Furnace fuel gas must be control by a ratio of effluent flowrate and the temperature desired [1], a shut off valve must be present in case of fuel leakage.

Quench tank can operate at higher temperatures than the first reactor, since it's made out stainless steel, which is designed to undergo high temperatures.

3.1.5 Potential Disturbances

Malfunction of any of the controls installed can cause a disturbance in the system and jeopardize the production of vinyl chloride. A higher feed flowrate than the desired can result in an overflow of the reactor, which can result in a high waste of material. A bad temperature controller can result in pressure buildup inside the reactor, in case there is also a failure with the safety pressure valve, an explosion can occur. A possible fuel leakage going into the pyrolysis furnace is another concern in the plant.

In case hydrochloric acid is not totally dry, it can corrode the equipment and pipelines. Sometimes corrosion is only noticeable when leakage starts to take place, which would result in a shutdown of production for maintenance.

3.2 Right Side of Process

Figure 3 below shows the right side of the Vinyl Chloride process. This contains the two distillation columns, the HCL column and the Vinyl Chloride Column. It also includes streams 9-15 which will be analyzed to determine Materials of Construction.

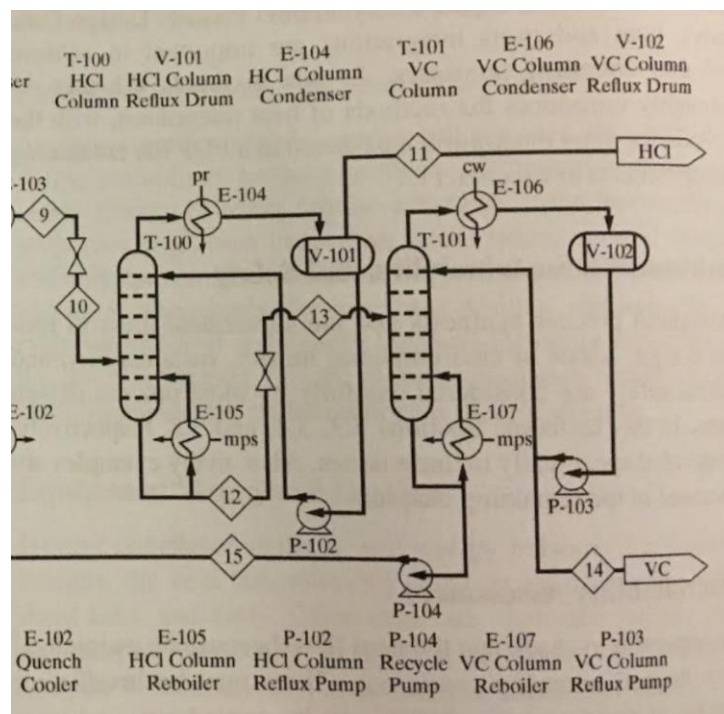


Figure 3: Right Side of The Process Before Heat Integration

3.2.1 Materials of Construction

Table 3: Properties of stream 9 to 15.

	Stream 9	Stream 10	Stream 11	Stream 12	Stream 13	Stream 14	Stream 15
Temp (°C)	6	6.5	-26.4	94.6	57.7	32.2	145.6
Pressure (atm)	26	12	12	12	4.8	4.8	4.8
Molar flow (lbmol/hr)							
1,2-dichloroethane	1,067	1,067	0	1,067	1,067	0	1,067
Vinyl Chloride	1,600	1,600	0	1,600	1,600	1,600	0
Hydrogen Chloride	1,600	1,600	1,600	0	0	0	0

Stream 9 and Stream 10:

This stream contains 1,2-dichloroethane, Vinyl Chloride, and Hydrogen Chloride. Both stream 9 and 10 have the same molar flow rates of the components and very similar temperatures. These streams will use the same metal. Hydrogen Chloride is very corrosive and will react with aluminum alloys, iron and steels. 1,2-dichloroethane is not compatible with aluminum, magnesium, and zinc and their alloys [4]. Vinyl Chloride isn't compatible with copper and aluminum [5]. All these compounds have very limited materials that are they are compatible with but the most affordable metal that would work for all of them would be Nickel alloys. Incoloy and Inconel would be these best alloys to use because it doesn't contain copper. Incoloy is cheaper to use and it is better for streams that don't have extreme conditions. These streams have relatively low temperatures and pressures so Incoloy would be the most efficient metal to use for these streams.

Distillation Column 1 (HCL Column):

The stream that enters this column is stream 10. Since stream 10 contains vinyl chloride, 1,2-dichloroethane, and hydrogen chloride, the material chosen for this unit must be compatible with these 3 components. The temperature and pressure entering the column from stream 10 aren't extreme so no extreme measures need to take place. The material used for stream 10 is Incoloy and this would also be the best material to use for this first distillation column. It will be able to prevent corrosion the best from the hydrogen chloride.

Stream 11:

This stream only contains Hydrogen Chloride at -26.4°C . Hydrogen Chloride is compatible with Aluminum bronze, Lead, Nickel-copper alloys, Platinum, Tantalum, Titanium, and Zirconium at temperatures lower than 20°C . Since there is so many options to choose from for this stream, the cheapest metal should be used. Out of that list Aluminum bronze is the cheapest metal to use.

Stream 12 and Stream 13:

Stream 12 and stream 13 both contain the same molar flow rates of 1,2-dichloroethane and Vinyl Chloride. The temperatures are slightly different but not extreme enough to differ the metal needed for each stream. Since hydrogen chloride isn't in this stream, steel is able to be used for the material of construction. The cheapest steel to use is Carbon Steel and Carbon Steel is extremely compatible with both 1,2-dichloroethane and Vinyl Chloride at these temperatures [6]. It is the most affordable and commonly used so it would be most efficient to use CS for streams 12 and 13.

Distillation Column 2 (VC Column):

Stream 13 is the stream that is entering the second distillation column which is named the Vinyl Chloride Column. This stream contains 1,1-dichloroethane and vinyl chloride, so the material used must be compatible with these two compounds. Just like stream 13, the best most affordable material to use would be Carbon Steel for the column. This would be the best economical decision and CS has excellent compatibility for these two compounds.

Stream 14:

Stream 14 contains a molar flow rate of 1,600 lbmol/hr of Vinyl Chloride and this is the only component in this stream. The metals that are compatible for vinyl chloride are considered for this stream. Vinyl chloride is compatible with most commonly used metals except for copper and aluminum. Carbon steel is the cheapest metal out of all the compatible metals for vinyl chloride.

Stream 15:

This stream only contains 1,2-dichloroethane at 145.6°C. Carbon steel would make the most economic sense for this stream as well because it is compatible with this compound and it is able to withstand a temperature of 145.6°C.

3.2.2 Constraints

In the second distillation columns the VC is produced in the overhead and must contain about 99.9 wt%. In order to make VC this pure, there will need to be some process controls implemented into the process.

3.2.3 Control Objectives

The objective for the first distillation column is to use a distillation process to separate HCl from Vinyl chloride and 1,2-dichloroethane. This HCL is the overhead product from the column and can be recycled to the oxychlorination reactor. The bottoms product consists of VC and 1,2-dichloroethane and the second distillation column is used to separate the VC from the 1,2-dichloroethane.

Table 4: Variables

Controlled Variables	Manipulated Variables
Pressure in columns	Condenser pressure control
Level in columns	Outlet valve
Composition of the tops products	Tray temperature control
Vapor-liquid level in reflux drum	Level control of the reflux drum
Vapor-liquid level in reboiler	Flow control of the steam

Figure 4 below is the P&ID for the right side of the vinyl chloride process. For distillation columns there are usually 5 degrees of freedom when it comes to process control. The pressure in the column must be controlled because if it exceeds a certain pressure there can be many safety concerns. This can be controlled with a condenser pressure control which controls the coolant flow into the condenser. The level in the columns must be controlled to prevent overflowing of the column and an outlet valve can be used for this. The composition of the tops product for both columns is the most important objective. It must be ensured that this composition is controlled throughout the process. Tray temperature is the manipulated variable because tray temperature can infer the composition. This inferred composition can then adjust the reflux rate to make sure the composition is as purified as can be. The vapor-liquid level in the reflux drum and in the reboiler must be controlled. Since the tops product is most important to purify, the reflux drum can use a level control. The bottoms reboiler will use a flow control to control the flow of the steam into the reboiler.

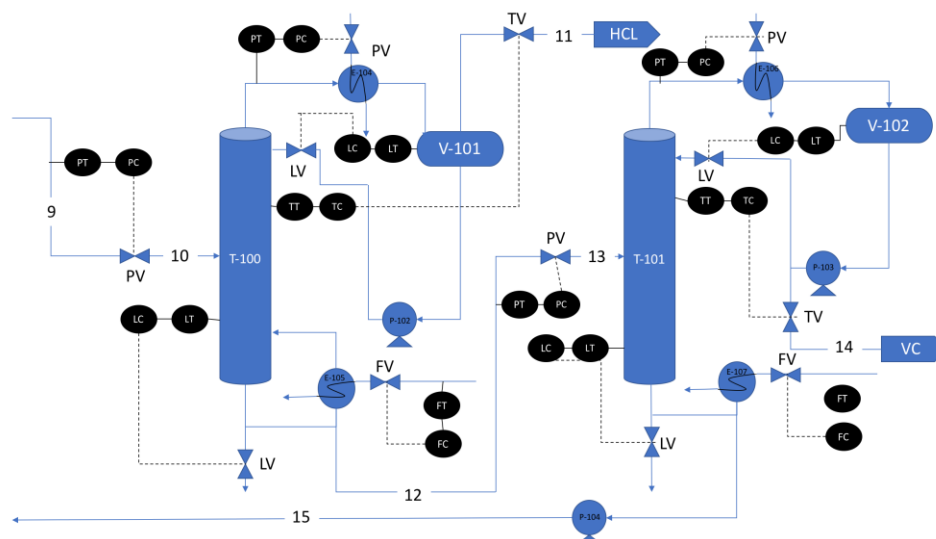


Figure 4: Process Control Right Side

3.2.4 Safe Limits for Process Variables

The first distillation columns operate at around 9 atm so the pressure control should control this column to around that pressure or else there can be safety issues [1]. The second column is constructed of carbon steel, so the temperature cannot exceed 480°C because carbon steel can't handle temperature higher than that. The temperature also can't go below 45°C.

3.2.5 Potential Disturbances

The main objective for distillation columns is to control the compositions of the top and bottom products. These controls implemented in this process control must fix any disturbances that may occur. Steam and hot utility are a potential disturbance and the process controls must control the flow rate to the reboiler to make sure it is operating at the desired temperature. Cooling water is another potential disturbance to this process and must be maintained at the correct flow or temperature to keep this process operating correctly.

4.0 After Heat Integration

In this section it will be presented the production of vinyl chloride after the heat integration takes place. The analysis will be divided between the left and right side of the process.

4.1 Left Side of Process

Figure 5 below shows the left side of the Vinyl Chloride production after heat integration. This portion of the process contains a direct chlorination reactor, a pyrolysis furnace and a quench tank.

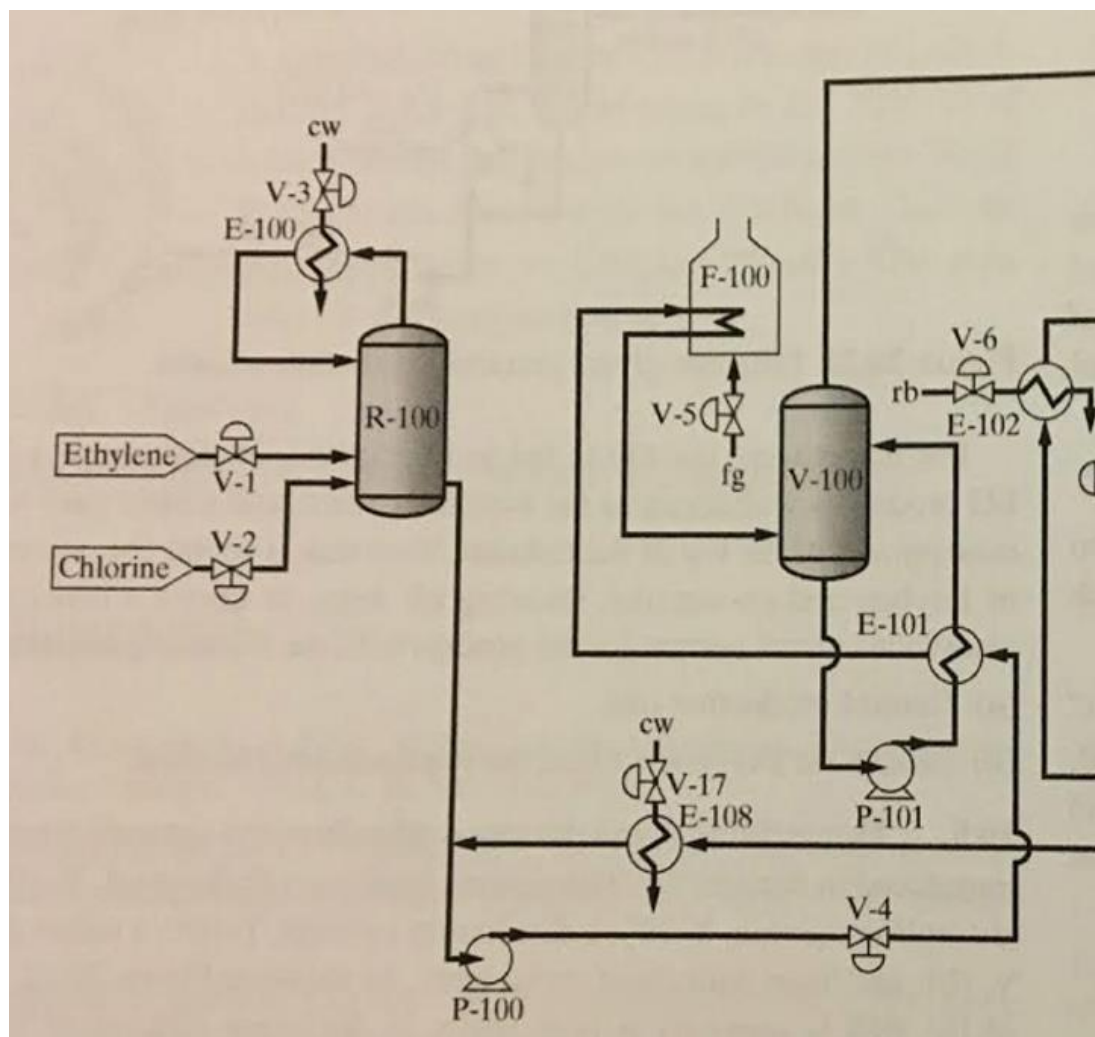


Figure 5: Left side of the production of vinyl chloride after heat integration

4.1.1 Materials of Construction

Table 5: Properties of streams 1 to 8.

	Stream 1	Stream 2	Stream 3	Stream 4	Stream 5	Stream 6	Stream 7	Stream 8
Temp (°C)	25	25	90	90	91.3	242	500	170
Pressure (atm)	1.5	1.5	1.5	1.5	26	26	26	26
Molar flow (lbmol/hr)								
Ethylene	1600	0	0	0	0	0	0	0
Chlorine	0	1600	0	0	0	0	0	0
1,2 - dichloroethane	0	0	1600	2667	2667	2667	1067	1067
Vinyl chloride	0	0	0	0	0	0	1600	1600
Hydrogen chloride	0	0	0	0	0	0	1600	1600

Although the streams in the after-heat integration haven't been numbered, after analyzing the system it's possible to determine that only previous stream 5 changed position when compared with the before heat integration PFD, however, all the streams in this part of the process still have the same composition and physical properties as previously discussed and can be summarized once again in Table 5. For further details on the streams, please refer to Section 3.1.1. Due to the streams being the same as previously explained, the material of construction will also be the same, which means that the reactor shell will be out of carbon steel due to the temperature of the reaction and the corrosion properties of this material. Also, stainless steel tubing will be necessary due to the high temperatures in the process to certify a dried hydrochloric acid, preventing the corrosion of the material as discussed in the previous section.

4.1.2 Constrains

As explained in Section 4.1.1, the material of the equipment, as well as the physical properties and composition of the streams didn't change in between before and after heat integration. This will result in the same set of constraints as previously discussed in Section 3.1.2, please refer to previous section for details.

4.1.3 Control Objectives

Table 6: Control Variables for left side of the process

Equipment	Controlled Variable	Manipulated Variable
Feed flowrate	Chlorine flowrate	Ethylene Flowrate
1 st Reactor	Temperature of reactor	Colling water flowrate
1 st Reactor	Pressure of Reactor	Safety Pressure release Valve
Pyrolysis Furnace	Inside temperature of the furnace	Fuel gas flowrate
2 nd Reactor	Temperature of the Reactor	Pump 101 Power

As Table 6 indicates, there are some controlled variables for the left side of the vinyl chloride production after heat integration. Starting from the feed flowrate, this system control will make sure that the flowrate going into the reactor is in a proper ratio to maximize the production of vinyl chloride. If the flowrate of one of the feed streams change, the flow transmitter will emit a signal to the flow indicator control to manipulate the valve responsible for the flowrate.

The second controlled variable is the temperature of the first reactor. The reactor is designing to operate under 100°C, and to make sure this is the case a cooling jacket was installed around it. The temperature element will measure the inside temperature of the reactor, and communicate it to the temperature transponder, which will send a signal to the temperature controller to adjust the temperature valve accordantly with the needs of the reactor. This valve controls the cooling water flow used to cool the jacket's fluid. By increasing the flowrate of the water, a higher heat exchanger will be created between the fluid and the water, cooling the reactor down. In case of decrease of the water flow, a higher internal temperature will be obtained inside the reactor.

Just like in the scenario before heat integration, a safety pressure released valve was installed as a prevention of a completely failure of the temperature control. In case that happens, the pressure transmitter will communicate to the pressure indicator control, which will control the pressure valve to operate as needed.

In the pyrolysis furnace, the same control system was maintained when compared to the before heat integration scenario. To reach the desired temperature, a temperature element will measure the outlet stream, communicating it to the temperature transponder, which will send a signal to the temperature control to manipulate the fuel gas valve, names temperature valve, as needed. However, a pre-heated stream 5 is now been used as a heater exchanger for the quench tank, by absorbing the heat been released by the jacket's fluid of the vessel.

For the quench tank the control variable changed from being the flow of cooling water to be the power of the pump P101. In order to reach the temperature desired in the outlet of the quench tank, a cooling jacket was installed around the vessel. The cooling fluid exchanges heat with stream 5, however, the pump determines the flow of cooling fluid circulating through the jacket, which results in the temperature desired. By installing a temperature element at the outlet vessel stream, the temperature transponder can send a signal to the temperature control

to adjust the flow of the fluid accordingly with its needs, which will change the power of the pump been operated. A visual summary of this discussion can be seen at Figure 6.

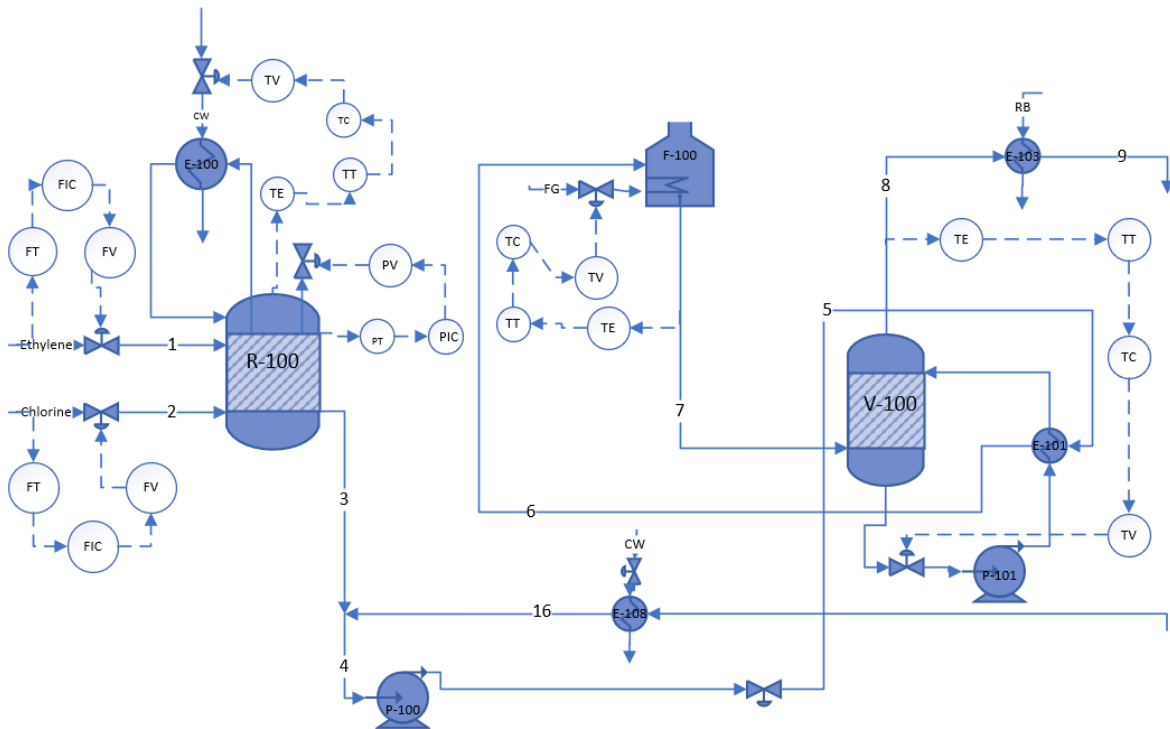


Figure 6: Left side of the vinyl chloride production after heat integration.

4.1.4 Safe Limits for Process Variables

Safe limits for process variables are the same as described at Section 3.1.4. Please refer to previous section for details.

4.1.5 Potential Disturbances

Potential disturbances are the same as described at Section 3.1.5, only with the addition of the pump for the quench tank. In order to avoid any disturbances, the flowrate going through the pump must be steady, to avoid any air bubble to damage the pump.

4.2 After Heat Integration Right Hand Side of Process

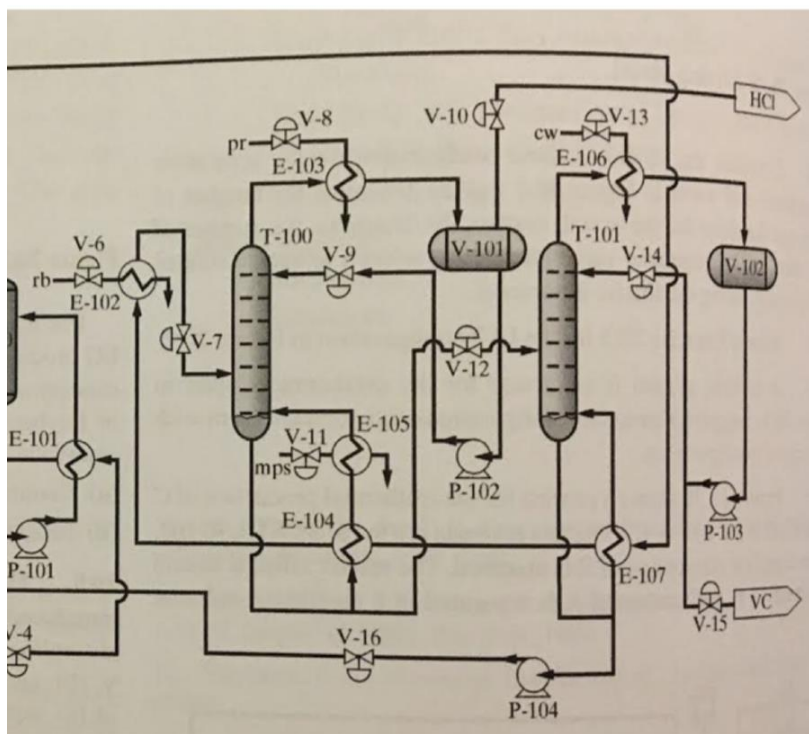


Figure 7: Right-Hand Side After Heat Integration

4.2.1 Materials of Construction

Table 7: Properties of Streams 9-15

	Stream 9	Stream 10	Stream 11	Stream 12	Stream 13	Stream 14	Stream 15
Temp (°C)	6	6.5	-26.4	94.6	57.7	32.2	145.6
Pressure (atm)	26	12	12	12	4.8	4.8	4.8
Molar flow (lbmol/hr)	X	X	X	X	X	X	X

1,2-dichloroethane	1,067	1,067	0	1,067	1,067	0	1,067
Vinyl Chloride	1,600	1,600	0	1,600	1,600	1,600	0
Hydrogen Chloride	1,600	1,600	1,600	0	0	0	0

Stream 9 & Stream 10

These two streams contain Vinyl Chloride, Hydrogen Chloride, and 1,2-Dichloroethane. Out of these three compounds, hydrogen chloride is the most corrosive. Its corrosive properties are much harsher on metals, not including copper, and lighter on plastics such as PVC and polytetrafluoroethylene (PTFE). Stated previously, these three compounds have limited choices for piping construction material. 1,2-dichloroethane and vinyl chloride is also very compatible with PTFE. Previously chosen was a nickel-based Incoloy piping system. The stream is at a relatively low temperature and pressure. For after the heat integration, PTFE lined steel is the best choice and very compatible with all three compounds.

1st Distillation Column

Flow into the first distillation column is stream 10, which contains vinyl chloride, 1,2-dichloroethane, and hydrogen chloride. All three are found in stream 10, so it would be best to use the same material. For the distillation column, the nickel-based Incoloy would be the ideal material to use.

Stream 11

Hydrogen chloride is the one and only compound found in this stream. A nickel-based alloy could be used again. However, the cost/piping ratio for PTFE is much more favorable than a nickel alloy of the same length. PTFE would be the ideal choice for piping material for stream 11.

Stream 12 & Stream 13

These two streams contain 1,2-dichloroethane and vinyl chloride. Carbon steel provides good resistance to corrosion for both compounds. Being both affordable and highly resistant, carbon steel is the ideal choice for stream 12 and 13 piping.

2nd Distillation Column

Flow into the second distillation column is the same as 13, vinyl chloride and 1,2-dichloroethane. As with stream 13, carbon steel provides good resistance to corrosion for both compounds. With carbon steel also having a good cost to piping length ratio, it would be the ideal choice for piping material for the second distillation column

Stream 14

This stream only contains vinyl chloride. The temperature is lower than stream 12 and 13, while the flowrate did not change. As with the previous two streams, carbon steel would be the ideal choice for piping for stream 14.

Stream 15

This stream only contains 1,2-dichloroethane at 145.6 °C. As with the previous three streams, carbon steel would be the ideal choice for piping for stream 15.

4.2.2 Constraints

The first distillation column is made from a nickel-alloy. Its high versatility combined with its outstanding heat and corrosion resistance. An example would be Hastelloy B, which is known for its resistance to hydrogen chloride and for its creep and rupture strength at temperatures around 760°C [7]. The second column is made from carbon steel. It can withstand temperatures up to 480°C and pressure from 160 to 200 psig.

4.2.3 Control Objectives

Table 8: Control variables

Controlled variables	Manipulated Variables
Temperature of Flow	Flow Valve
Pressure in columns	Pressure Release Valve
Reflux Ratio	Level Controller
Reboiler Temperature	Temperature Control valve

The bottom control loop uses a steam flow rate controlled by a level controller. The differential pressure controller has a set point equal to the maximum drop in tray pressure and the set point of the level controller is the desired bottoms composition. A level controller on the reflux drum is used by the overhead control loop to control the reflux ratio.[1]. The temperature at the bottoms and distillate is regulated by a temperature controller. The bottoms temperature is controlled by the steam flow rate to the reboiler. An additional temperature controller is located at the release valve before the first column. The distillate composition is the one needed to be controlled directly, as such a level controller is needed after the condenser to control the reflux.

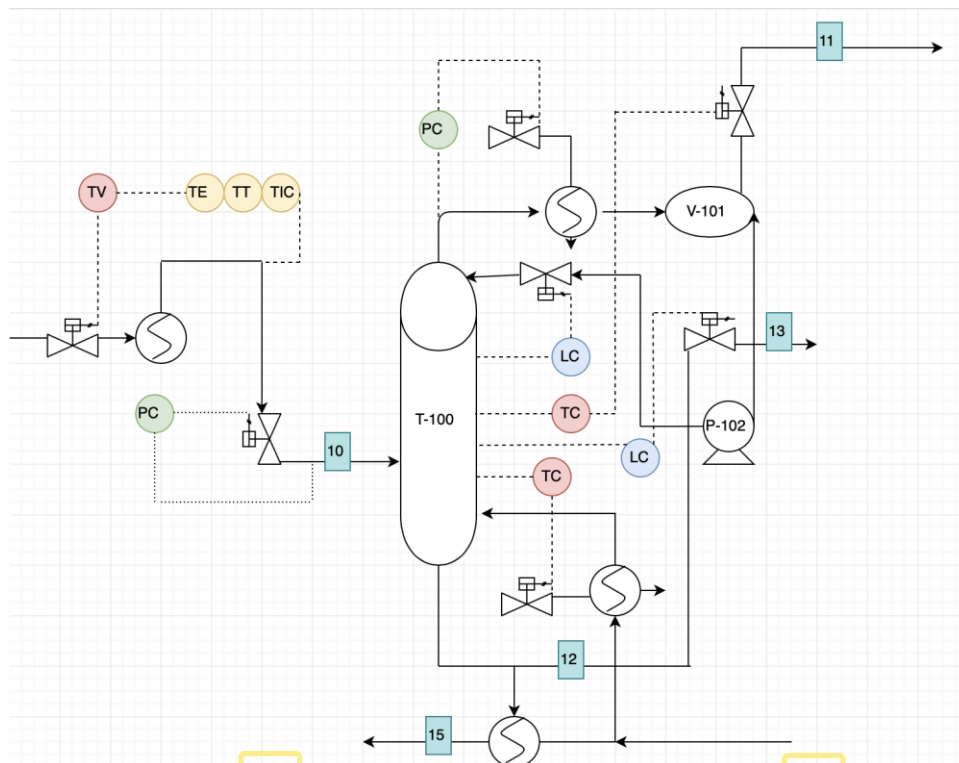


Figure 8: First Distillation Column's Control System

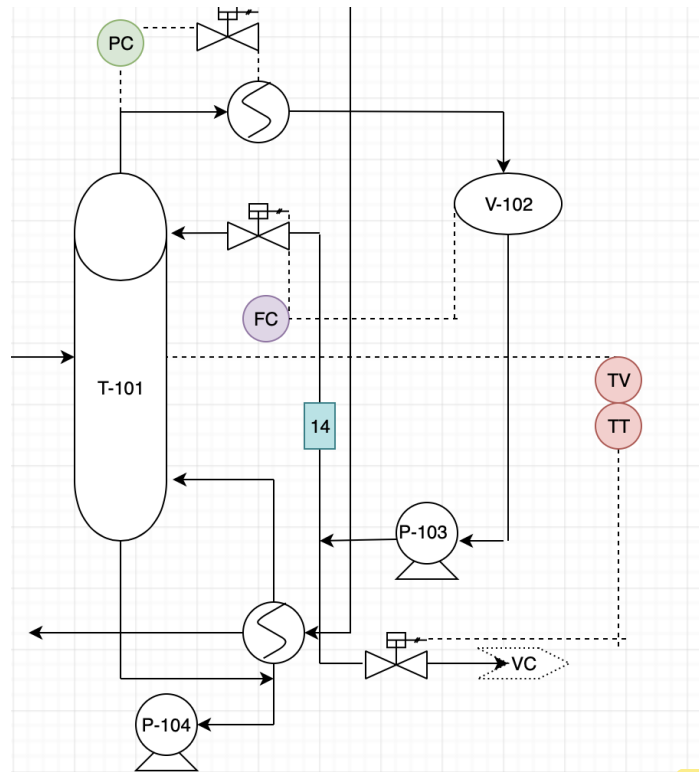


Figure 9: Second Distillation Column's Control System

4.2.4 Safe Limits for Process Variables

Safe limits of operation is set between 60psig and 135 psig, with a column 10psig pressure drop. Operating temperature should be around 315°C [7]. Temperature limit has no lower limit, but the system cannot exceed 500°C set by as a safety limit. Although though the melting point of carbon steel is around 1400°C, its best to have an alarm set. Flow Rates of incoming feed must also be at a reasonable level as to decrease stress on the system

4.2.5 Potential Disturbances

Potential disturbances are the same as described at Section 3.2.5, with the addition of the multiple utilities for a hat integrated system. Hot utility using steam are a potential disturbance and the process controls must control the flow rate to the reboiler to make sure it is operating at the desired temperature. Cooling water is another potential disturbance to this process and must be maintained at the correct flow or temperature to keep this process operating correctly.

5.0 Conclusions of the Process

The definition of feedback is a system which its controls measure an output variable and reacts by manipulating the controlled variable as needed. On the other hand, feedforward is the type of control where the disturbance of the system is measured, and the controlled variable is manipulated in order to avoid the disturbance in the system. The production of vinyl chloride process makes use of the feedback control system, by measuring a controlled variable and by adjusting the system accordingly to its needs.

This process is a continuous process, which makes sense to have a feedback control system to adjust any variable as needed during the production, which can include change of flowrate, change of temperature, change of pressure and change of reflux in the distillation column.

6.0 References:

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